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Employing Machine Learning for Seismic Intensity Estimation Using a Single Station for Earthquake Early Warning

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Abstract: An earthquake early-warning system (EEWS) is an indispensable tool for mitigating loss of life caused by earthquakes. The ability to rapidly assess the severity of an earthquake is crucial for effectively managing earthquake disasters and implementing successful risk-reduction strategies. In this regard, the utilization of an Internet of Things (IoT) network enables the real-time transmission of on-site intensity measurements. This paper introduces a novel approach based on machine-learning (ML) techniques to accurately and promptly determine earthquake intensity by analyzing the seismic activity 2 s after the onset of the *p*-wave. The proposed model, referred to as 2S1C1S, leverages data from a single station and a single component to evaluate earthquake intensity. The dataset employed in this study, named "INSTANCE," comprises data from the Italian National Seismic Network (INSN) via hundreds of stations. The model has been trained on a substantial dataset of 50,000 instances, which corresponds to 150,000 seismic windows of 2 s each, encompassing 3C. By effectively capturing key features from the waveform traces, the proposed model provides a reliable estimation of earthquake intensity, achieving an impressive accuracy rate of 99.05% in forecasting based on any single component from the 3C. The 2S1C1S model can be seamlessly integrated into a centralized IoT system, enabling the swift transmission of alerts to the relevant authorities for prompt response and action. Additionally, a comprehensive comparison is conducted between the results obtained from the 2S1C1S method and those derived from the conventional manual solution method, which is considered the benchmark. The experimental results demonstrate that the proposed 2S1C1S model, employing extreme gradient boosting (XGB), surpasses several ML benchmarks in accurately determining earthquake intensity, thus highlighting the effectiveness of this methodology for earthquake early-warning systems (EEWSs).

Keywords: on-site intensity; earthquake early warning (EEW); peak ground acceleration (PGA); machine learning (ML); synthetic ground motion

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1. Introduction

Earthquake early warning (EEW) plays a crucial role in mitigating the risk posed by natural disasters to human life. Extensive research has been conducted to develop strategies for reducing the impact of earthquakes (EQs) by leveraging various modern technologies like the Internet of Things (IoT) and others [1–5]. Social networking platforms have also been explored as a means to decrease the vulnerability to EQ disasters. Avvenuti et al. [6] combined social media capabilities with traditional methods to mitigate the impact of disasters. Furthermore, they have expanded their studies to incorporate remote sensing techniques supported by satellite imagery, along with the implementation

of software-defined network technology and IoT on-the-fly gateways [7–11]. Significant efforts have been made to assist regions with severely damaged infrastructure, and virtualization has emerged as a valuable tool for redirecting disastrous risk assessment situations. Bao et al. [12] integrated a diverse virtual network and cloud system to simulate a catastrophic scenario for an early-warning system (EWS), facilitating secure evacuation. Remote sensing techniques have also played a significant role in enhancing EQ resilience and reducing urban damage [13–15]. Such studies can contribute to building seismic resilience and mitigating urban damage [16–19].

These initiatives are recommended as supplementary approaches to the existing methods for fault characterization outlined in Brune's work [20] and for detecting EQs. Li et al. [21] proposed a model that has an analogy in the mechanism to assess the coherence of received signals. Other studies, like those by Olson and Allen [22] and Kanamori [23], focused on estimating EQ magnitude using data from the initial seconds of the rupture. The initial seconds after an EQ are crucial for EEW due to the relationship between the blind zone and the monitored window. A larger blind zone, which reserves time before powerful EQ waves arrive, results in a longer predicted window size.

The complexity of seismic waves and geological composition provides a direction for the development of intelligent and reliable solutions. Modern technologies, like ML, have the potential to significantly reduce the impact of EQ disasters [24–27]. ML is an intelligent technique that can be utilized to resolve complex issues without relying on specific mathematical approaches [28]. It has been widely used for tasks like missing data reconstruction and data analysis. In the literature, many ML models have been employed in various credit scoring models [29–32]. However, there are instances where conventional statistical analysis methods may lack effectiveness, as certain assumptions made by these models are unverifiable, which can affect the accuracy of predictions. Advanced ML methods like support vector machines (SVMs), Naïve Bayes (NB), and random forests (RFs) have demonstrated superior performance compared to statistical models. Various measures can be employed to assess the performance of different ML models, as discussed in Hossin and Sulaiman's review [33]. For the development of an effective EEWS, it is crucial to analyze multiple aspects, including EQ location and magnitude. Precise and timely estimates of parameters like peak ground acceleration (PGA) leading to the on-site intensity can greatly contribute to saving human lives [34,35].

1.1. Motivations

In many studies within the literature, approaches relying on extensive feature sets often encounter challenges in detecting gaps, managing high complexity, and discerning non-linear patterns within observed waveforms. For instance, the researchers in [36] devised two novel locking models to scrutinize the PGA hypothesis within dynamic rupture scenarios, exploring different ground-motion models alongside real-world events. Additionally, the exploration of EQ early-warning systems (EEWS) for future deployment was addressed in [37], where observations of position, magnitude, and rupture length formed the foundation of the proposed model.

Further advancements were made in [38], where recurrent neural networks (RNNs), specifically long short-term memory (LSTM) models, were employed. These models, coupled with a fusion of ground-motion intensity data, contributed to the development of a comprehensive framework aimed at understanding ground motion associated with subduction interface events and forecasting PGA. Despite the multitude of endeavors documented in the existing literature, a notable gap persists. Specifically, the utilization of weak-motion data, particularly within a narrow 2 s window following the *p*-wave, has yet to be explored or suggested. This represents a potentially fruitful avenue for future research in seismic analysis and early-warning systems.

The utilization of weak-motion data within a narrow time window following the *p*-wave holds significant potential for improving EEWSs and enhancing our understanding of ground-motion dynamics. This approach addresses the limitations of methods relying

solely on strong-motion data, which often struggle to capture gaps, high complexity, and non-linear patterns in seismic waveforms.

Incorporating weak-motion data offers several advantages for EQ analysis. By capturing seismic signals with lower amplitudes, weak-motion data provides additional insights into the early stages of an EQ event. This allows for a more comprehensive understanding of its characteristics and behavior, including the detection of complex rupture scenarios and non-linear patterns that may not be evident in strong-motion data alone. The early stages of an EQ contain critical information about the magnitude and intensity of the event. By leveraging weak-motion data in combination with advanced machine-learning techniques, like RNNs and LSTM models, it is possible to develop robust predictive models that effectively forecast PGA and inform early-warning systems.

The proposed approach of utilizing weak-motion data within a limited time window following the *p*-wave has practical implications for EQ hazard mitigation and risk reduction. By expanding the scope of data collection to include weak-motion signals, we can enhance the EEWS robustness. This, in turn, enables more effective response strategies, like timely evacuations, structural reinforcements, and emergency preparedness measures, ultimately reducing the potential impact on human lives and infrastructure. It is worth mentioning that using ML for on-site intensity estimation is crucial as it enhances accuracy, speed, and reliability compared to traditional methodologies, allowing for more precise and timely EEWSs.

1.2. Contributions

This work introduces an EEWS using an ML approach to classify on-site intensity. The system, called 2S1C1S, incorporates various ML models to analyze faint motion waveforms. The key idea is to use just a 3 s window of the waveform (1 s before the *p*-wave and 2 s after), simplifying implementation and ensuring the effectiveness of the EEWS. By extracting relevant features from the seismic waveform, the system can make fast and reliable decisions using ML techniques. The main contributions of this paper are as follows:

- A new classification method categorizes EQ intensities within 3 s (1 s before and 2 s after the *p*-wave), achieving 99.05% accuracy using single-component–single-station data. The model uses a dataset of 27,531 records from 386 INSN seismic stations.
- Two different conversion methodologies from the on-site PGA to the corresponding on-site intensity are employed. The USGS developed one [39] and the other was developed by Faenza et al. [40].
- The method is cost-effective, relying only on on-site velocity waveforms, eliminating the need for both strong- and weak-motion stations, simplifying data collection, and optimizing resources while accurately gauging EQ intensity in real time.
- Various ML techniques are used to identify the most effective for distinguishing EQ intensities, enhancing the EEWS with thorough risk evaluation and activity categorization, and minimizing incorrect hazard evaluations. Five ML methods are evaluated and compared using various metrics.
- Integration with IoT systems is recommended for reliable, swift alerts to decision-makers, essential for an effective EEWS and evacuation plans to mitigate EQ impacts.

This paper is structured as follows: Section 2 provides an overview of related work, Section 3 provides details about the experimental setup and data used, Section 4 describes the five ML approaches and the proposed system model, Section 5 outlines the evaluation criteria for determining the optimal ML model, and Section 6 presents the obtained outcomes. Section 7 presents a discussion about the proposed approach among the related works and, finally, the paper is concluded in Section 8.

2. Related Work

In this section, we provide an overview of related work in the field of EEWSs and the use of ML techniques for monitoring and detecting EQ parameters. Several studies have focused on utilizing ML and IoT technologies to enhance EEWSs and risk management in

hazardous settings. For example, Poslad et al. [41] proposed a data-driven IoT system for EEWSs that addressed scalability and data exchange challenges. The system utilized small IoT nodes for efficient data transmission and real-time monitoring [42,43].

Chung et al. [44] investigated strategies for efficient data sharing in EEWSs using IoT connectivity. They employed web application nodes and message queuing telemetry to facilitate data exchange and mitigate the impact of disasters. These studies highlight the importance of IoT and data-driven approaches in developing robust and scalable EEWSs.

In terms of EQ parameter detection, various ML techniques have been explored. Wu et al. [45] presented a convolutional neural network (CNN) model for EQ detection based on low-frequency acoustic energy. Their model aimed to locate and determine the magnitude of EQs using waveform analysis. Lomax et al. [46] proposed a deep learning technique for EEWSs that relied solely on data from individual station waveforms.

Different ML models have also been utilized to differentiate between EQ and noise records [47]. DeVries et al. [48] employed ML techniques to distinguish between an original EQ and its consequences. Chen et al. [49] proposed a deep learning denoising method for seismic data and detecting event arrival times. These studies demonstrate the application of ML in improving the accuracy and efficiency of EQ detection and classification.

The detection of the *p*-wave, which is very important for early-warning systems, has been a focus of several studies. Ross et al. [50] utilized CNNs to analyze *p*-wave detection, although a considerable amount of training data was required. He et al. [51] introduced a distinct DL model named PickCapsNet for identifying the *p*-wave arrival time. They achieved accurate *p*-wave detection using their model. Autoencoders (AEs) have also been employed in EQ waveform analysis. Mousavi et al. [52] utilized AE models for event arrival time detection and seismic data denoising.

In addition to EQ detection, other factors like PGA, peak ground velocity (PGV), peak ground displacement (PGD), and active tectonics play crucial roles in assessing seismic hazards [53–56]. Various studies have employed ML models to predict these parameters [57]. For instance, Yao et al. [36] used DL models to evaluate PGA predictions for dynamic rupture scenarios. Fayaz et al. [58] utilized DL models to estimate on-site ground acceleration and predict the acceleration response spectrum. Hsu et al. [59] employed an LSTM model to forecast on-site PGA by detecting order dependency in seismic waves. These studies highlight the potential of DL in predicting seismic parameters for EEWSs.

Efficient implementation of EEWSs requires the accurate identification of seismic phase types and propagation directions. Jung et al. [60] used ML techniques to determine the back azimuth and phase slowness of local events, enabling efficient event identification. Bose et al. [61] focused on fault rupture details to predict PGA and assess the extent of damage. They examined EQs in California, Oregon, and Washington to gain insights into seismic hazards.

To sum up, while there have been efforts to discriminate between different EQ intensities, there is no previous scheme published that relies exclusively on the proposed clarification criterion. The proposed approach in this paper aims to address this gap and provide a reliable and adaptable solution for EQ early-warning systems.

3. Data Preparation and Experiment Setup

In this section, we describe the data preparation process and the experimental setup used in this work. Our main objective is to measure on-site intensity using weak-motion data collected from multiple seismic stations. We begin by discussing the waveform collection and preparation process, followed by the intensity categorization and augmentation stages.

3.1. Dataset Statistics and Process

The dataset used in this study called INSTANCE is sourced from the INSN and is provided in HDF5 format [62], which is suitable for ML applications. The original dataset

consisted of approximately 1.200 million 3C traces obtained from around 50,000 data events. Each station, on average, contributed 21 events with 3C traces. The waveforms were sourced from the Italian Seismic Bulletin [63], a comprehensive repository covering the period from January 2005 to January 2020. The events in the dataset span a magnitude range of 0.0 to 6.5.

The recorded waveforms include strong-motion channels and weak-motion channels. Figure 1 illustrates the spatial distribution of events and station locations. All traces have a fixed duration of 120 s and were captured at a sampling rate of 100 Hz. To ensure consistency and comparability, the waveforms were transformed into physical units of ground motion by deconvolving the instrument transfer functions. This transformation allows us to utilize the ground-motion amplitude values, which are essential features for our analysis.

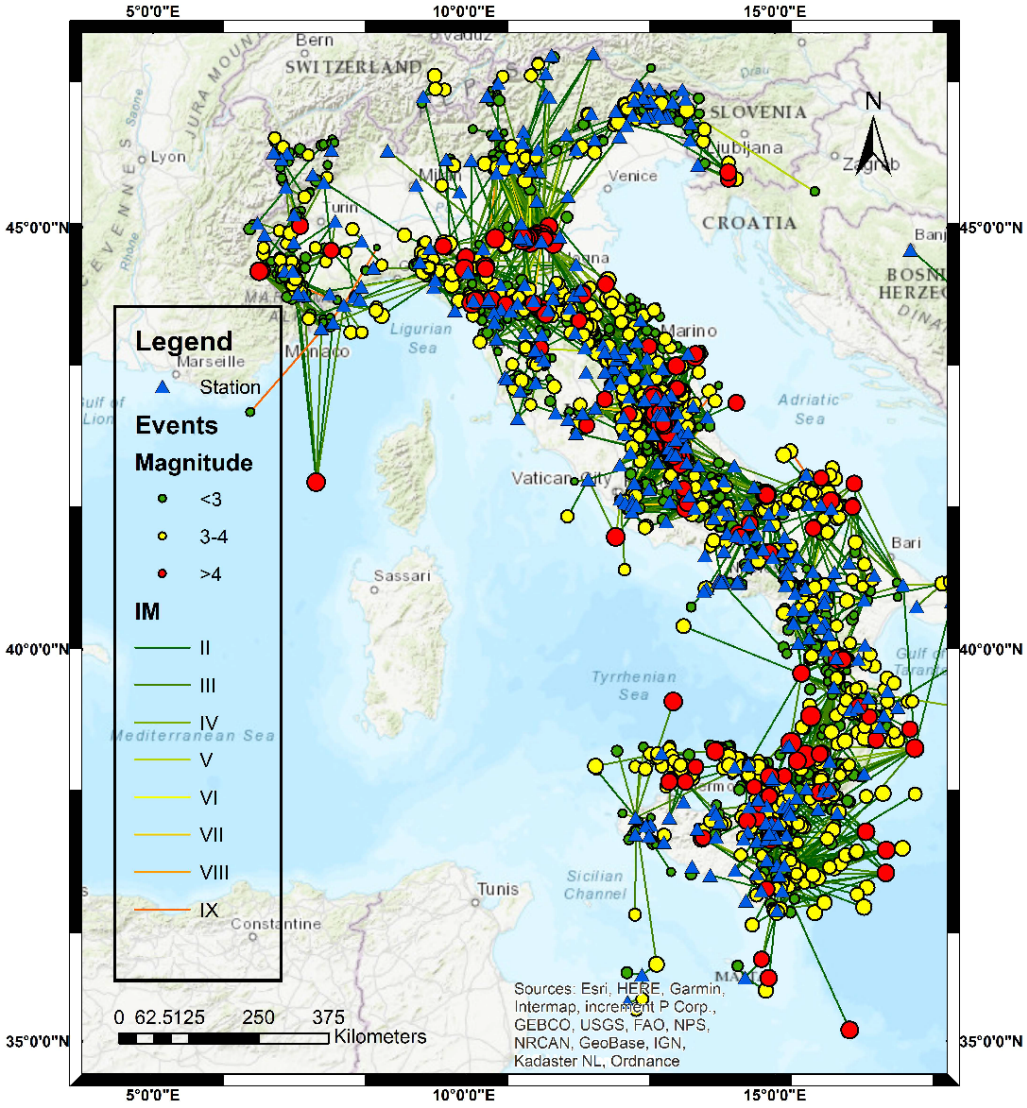


Figure 1. Event path ray, station locations, and intensities.

In addition to the waveform data, the INSTANCE dataset is enriched with metadata that provides detailed information about each event and the associated seismic stations. The metadata include over 100 parameters, like EQ location, station identification, trace properties, and derived values related to ground-motion intensity. These metadata parameters play a crucial role in our analysis.

According to [62], we utilized the dataset in units of physical ground motion, which had been prepared after deconvolving the instrument response. This preparation involved

using station response files for all the stations and applying transfer functions to the individual traces with frequency filtering corners at 0.01, 0.04, 25, and 40 Hz, using a cosine flank frequency domain taper and applying a 5% cosine tapering at both ends of the trace signal.

For each trace, a random angle between 0 and 360 degrees was used as a rotation angle. The trace waveform and the randomly selected angle were inputs for the rotation transform augmentation, producing a rotated version of the trace waveform. This operation was repeated several times for each trace to generate the required number of augmented traces. This rotation augmentation method generalized the dataset, ensuring varied data with each augmentation iteration. Swapping traces could be considered a rotation at an angle of exactly 90 degrees, with other azimuthal values implemented through rotation transform augmentation.

3.2. Intensity Categorization

To perform intensity categorization, we leverage waveform traces and various ground-motion metrics. These metrics include the optimal and minimal amplitudes of PGA and PGV, RMS value, and other ground-motion intensity measures. Several studies, like [40], have established empirical relationships between these metrics and the corresponding intensity levels.

In our analysis, we focus on the Modified Mercalli Intensity (MMI) scale, a widely used measure of EQ intensity. We aim to establish regression correlations between MMI and PGA, which is a commonly available metric in the dataset metadata. To achieve this, we perform a simple on-site conversion of PGA values to MMI values.

We utilize two regression relationships for the on-site conversion. The first relationship is based on USGS-developed regression correlations between MMI and PGA. This relationship was established by comparing the horizontal PGAs of eight recorded massive California EQs to evaluate on-site intensities [39]:

$$MMI = 3.66 \log(PGA) - 1.66. \quad (1)$$

The second relationship we employ is the one developed by [40], which is specific to the intended area by

$$MMI = \pm 0.14 \log(PGA) \pm 0.22 + 2.58 + 1.68, \text{for} = 0.35, \quad (2)$$

$$MMI = \pm 0.09 \log(PGA) \pm 0.07 + 2.35 + 5.11, \text{for} = 0.26. \quad (3)$$

In our study, we adopt the relationship developed by [39] as an example of intensity conversion. The proposed model is capable of incorporating any suitable relationship between MMI and ground motion.

After assigning a new intensity label (MMI value) to each 1C trace in the dataset, we exclude events with MMI falling into the I category. This exclusion is carried out to prevent any learning bias that may arise due to a large number of less significant events in this category. Figure 1 provides a graphical illustration of the spatial distribution of on-site intensities reported at each station.

3.3. Data Augmentation

Data augmentation is an important stage in our data preparation workflow. It entails creating extra training instances by performing various changes to current data. The purpose of data augmentation is to increase the diversity and robustness of the training set, which improves the ML models' generalization performance.

In the context of our study, data augmentation techniques are utilized to compensate for the lack of certain featured data. These techniques involve applying random rotations, translations, and scaling operations to the waveform data. By introducing such variations, we can enhance the model's ability to generalize to unseen data and capture the inherent variability in the seismic signals.

Specifically, for each waveform in the dataset, we apply random rotations within a certain range of angles, random translations in the time domain, and random scaling operations to alter the amplitude of the signals. These transformations are applied independently to each component of the 3C waveform, ensuring that the spatial relationship between the components is preserved.

By augmenting the dataset, we effectively increase the number of training examples available for the ML models. This reduces overfitting and increases the model's capacity to handle a variety of seismic events and recording settings. Additionally, data augmentation allows the models to learn robust representations that are invariant to certain variations in the input data.

The expanded dataset is then utilized to train and evaluate machine-learning models. It is vital to remember that the data augmentation process only applies to the training set, not the validation or test sets. This ensures that the models are evaluated using unseen, unaltered data, resulting in a more precise evaluation of their performance.

Figure 2a,b depict the histogram of Modified Mercalli Intensity (MMI) labels across the entire INSTANCE dataset. These histograms were generated after categorizing the labels based on the conversion equations established by [39,40], respectively, with the exclusion of instances labeled as "not felt". Notably, the visuals highlight an evident imbalance in label distribution across the classes, potentially introducing biases and inaccuracies in the model's learning process.

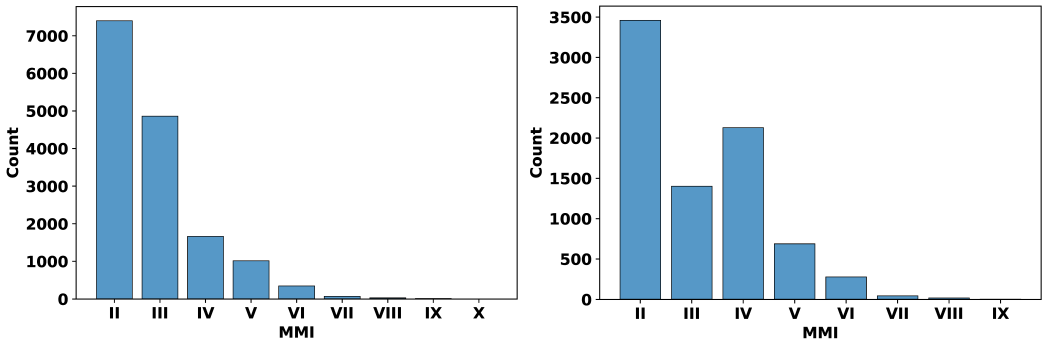


Figure 2. MMI label histogram via (**a**) equations developed by [39]; (**b**) equations developed by [40].

To address this issue, seismic waveform augmentation techniques were employed. However, it is important to note that intensity level I, denoting occurrences that are imperceptible or inconsequential, was omitted from the analysis as it likely represents a significant portion of the dataset and could exacerbate the imbalance issue. Additionally, due to their proximity in value and minimal impact, the weak intensity labels (II and III) were merged with the light intensity label (IV), resulting in the model being trained on six distinct labels. Moreover, training deep learning models on a limited dataset like INSTANCE raises concerns about overfitting. To mitigate this risk, data augmentation techniques were implemented to augment the volume and enhance the quality of the INSTANCE dataset. In seismology, common data augmentation methods include the addition of background noise, permutation, and cropping. While these procedures increase dataset quantity, they may not necessarily ensure dataset quality. To preserve data integrity, precise waveform transformations were applied, ensuring that any adjustments made were physically justifiable and maintained the dataset's fidelity.

To denote shifts in event locations while preserving the label integrity, we applied three distinct transformations to the seismic data. Firstly, we interchanged the North (N) and East (E) channels to alter the signal's azimuth. Secondly, we randomly rotated the waveform of the 3C data to a horizontal angle. Lastly, we flipped the polarity of the signal. These transformations aimed to simulate variations in seismic event paths without changing the associated labels. Additionally, we introduced further modifications to illustrate path

effects, like injecting random noise and using a Low Pass Filter with a randomly selected cutoff frequency. By combining these adjustments and applying each transformation type with a 0.5 probability, we generated synthetic data that closely resembles real-world seismic signals. This balanced dataset served as the foundation for training our model and enhancing the accuracy of seismological classification.

Following the augmentation process, adjustments were made to ensure the distribution of MMI classes remained balanced. Specifically, the number of classes ranging from III to X in Figure 3a and III to IX in Figure 3b was harmonized to match that of class II. This step was crucial in preventing potential overfitting issues. For the augmentation process, we leveraged a PyTorch library specifically designed for augmenting seismic data [64]. This resource facilitated the generation of the augmented INSTANCE dataset, ensuring its authenticity and reliability for training purposes. Figures 3 and 4 illustrate the correlation between MMI and depth, and MMI and distance, respectively. The input dimension refers to the number of features per waveform, 1 s before the commencement of the *p* wave, 2 s after it, and the scaling factor. The model processes 301 input characteristics per input sample, calculated as 3 s multiplied by 100 samples per second plus 1. The result is a one-hot encoded class used to categorize the MMIs, representing the real intensity classes.

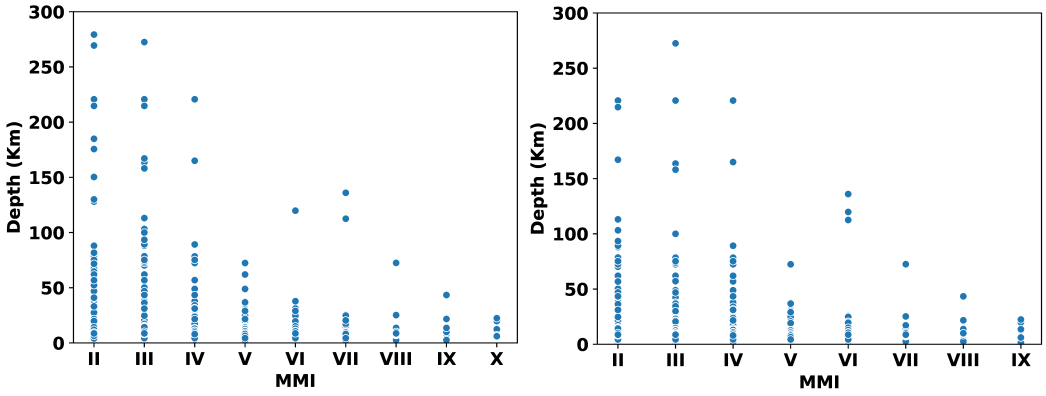


Figure 3. MMI vs. depth via (**a**) equations developed by [39]; (**b**) equations developed by [40].

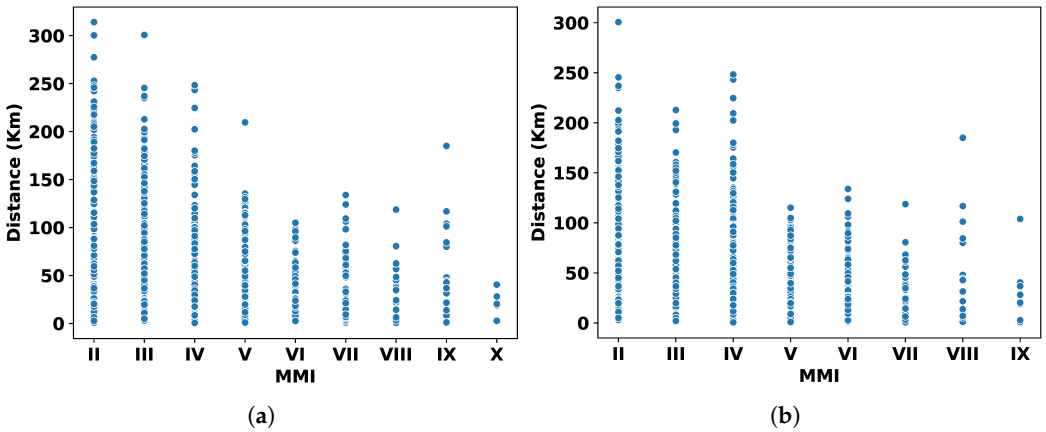


Figure 4. MMI vs. distance via (**a**) equations developed by [39]; (**b**) equations developed by [40].

3.4. Experimental Setup

In our experiments, we employ a discriminative approach to determine on-site intensity using the weak-motion data collected from multiple seismic stations. We divide the dataset into training, validation, and test sets to train, tune, and evaluate the ML models, respectively.

The dataset is randomly split into three parts: approximately 70% for training, 15% for validation, and 15% for testing. This partitioning ensures that the models are trained on a large enough set of examples, while still having sufficient data for validation and testing.

We use a variety of ML algorithms, including deep learning models, like CNNs and RNNs, as well as traditional machine-learning algorithms, SVMs, and RFs. These models are trained using the training set and tuned using the validation set.

To evaluate the performance of the models, we measure the accuracy of intensity prediction on the test set. Additionally, we calculate other metrics, like precision, recall, and F1 score, to assess the models' performance across different intensity categories. We also analyze the models' performance in terms of their ability to predict the intensity levels accurately for events recorded at different seismic stations.

To ensure fair comparison and reproducibility, we initialize the models with random seeds and use the same hyperparameters and training configurations for all the experiments. The models are trained using an appropriate loss function, like mean squared error or categorical cross-entropy, depending on the specific ML algorithm used.

4. Proposed Approach Using ML Classifiers

In this section, we present a proposed approach that utilizes ML classifiers for classifying EQ intensity. The approach incorporates a 3 s velocity wave. To capture the variability in real-world seismic events, a random shift in the *p*-wave arrival time within a range of ± 0.2 s is considered.

4.1. Extreme Gradient Boosting (XGB)

This model is a powerful gradient-boosting (GB) technique that outperforms traditional GB in terms of efficiency, dependability, and portability. It employs a large number of hyperparameters to achieve maximum accuracy and utilizes parallel tree boosting, which enables high-speed and efficient training [65]. XGB has gained significant popularity in recent years due to its excellent performance and versatility.

4.2. Decision Tree (DT)

The decision tree (DT) model is a non-linear classifier that can effectively handle complex problems by recursively partitioning the input space into smaller regions. Each partition is represented by a node in the tree, and the final classification decision is made based on the majority class in each region [66]. The decision tree's ability to divide problems into manageable chunks makes it a valuable tool for classification and regression tasks.

4.3. Random Forest

The random forest (RF) model is a type of ensemble learning that uses several decision trees to produce predictions. It addresses the limitations of individual decision trees by employing a coordinated ensemble of learners, which helps reduce variance and bias in the predictions [67]. The RF model aggregates the predictions of multiple trees and outputs the class label with the highest frequency as the final prediction. Mathematically, the prediction $\widehat{\mathbf{Y}}(\mathbf{f})$ of the RF model with NT trees is given by

$$\widehat{\mathbf{Y}}(i) = \frac{1}{NT} \sum_{j=1}^{NT} P_C(i), \tag{4}$$

where NT is the number of trees, the index of the input vector is i, and P_C represents the prediction of the classifier.

4.4. Extra Trees (ET)

The Extra Trees (ET) model is an extension of the RF model that offers a reduced risk of overfitting. It randomly selects a subset of features from the input dataset, INSTANCE, to support the decision-making process of the predictors [68]. By introducing additional randomness in the feature selection process, ET provides improved generalization capabilities and robustness against noise in the data.

4.5. K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) model is a non-parametric classification algorithm that assigns a label to an input sample based on the labels of its nearest neighbors in the feature space [69]. The decision boundary of the KNN model is determined by the majority vote of its nearest neighbors. The hyperparameter for this model is the number of neighbors (K) to consider, and the distance metric is used to determine the proximity between samples. KNN models can also take into account the weighted votes from the training set based on the cosine similarity to the input sample.

4.6. Proposed System Model

The model of our system, as shown in Figure 5, incorporates five ML models: extreme gradient boosting (XGB), DT, ET, RF, and KNN. The system model follows several steps to classify EQ intensity.

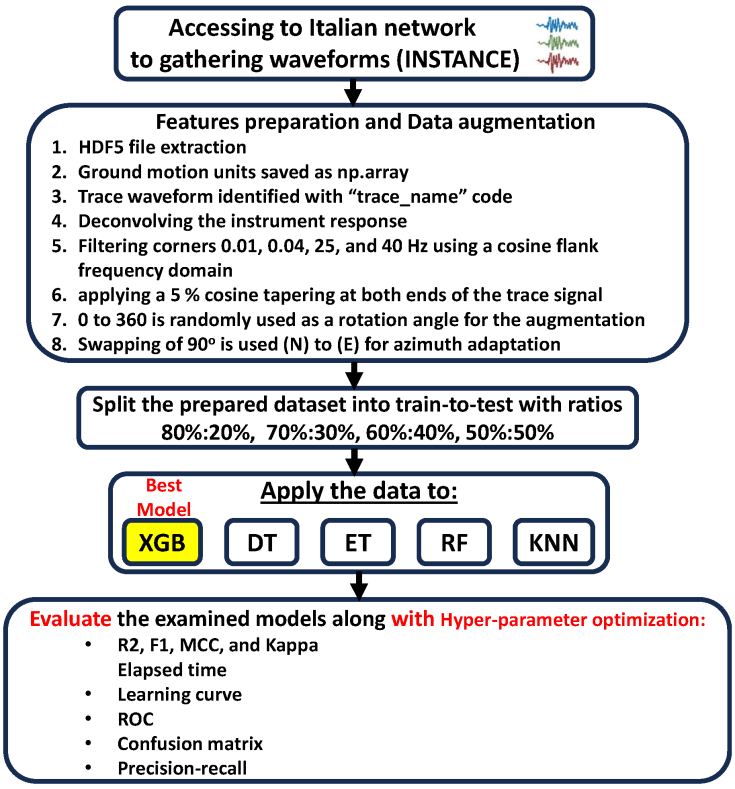


Figure 5. Proposed system model.

First, the system extracts the normalized sample amplitude and scaling factor from the given dataset INSTANCE. Then, a 3 s velocity wave is extracted, consisting of 1 s before the *p*-wave beginning and 2 s after to be fed into the ensemble of ML models.

To evaluate the models, the dataset is split into training and testing sets using four different ratios (50%:50%, 60%:40%, 70%:30%, and 80%:20%). Each split percentage is used to train and test the linear and non-linear ML models. The performance of the models is assessed based on various evaluation measures. The optimization step is performed using the four split percentages again, and the XGB model is identified as the top-performing model among the other models.

5. Evaluation Criteria

The ability to conduct considerably more detailed analysis than was previously possible due to advances in data storage and computer capability has helped to clarify how EQ intensities differ from each other. Consequently, we utilize the metrics presented below to assess classifier efficiency.

5.1. Proposed Model Accuracy

Accuracy is a commonly used metric to evaluate the performance of a classification model. It measures the proportion of correctly predicted targets relative to the total number of targets. In the context of EQ intensity classification, accuracy represents the percentage of correctly classified EQs.

The accuracy score is calculated as follows:

$$\text{Accuracy} = \frac{\sum_{i=1}^{N} I(y_{\text{obs}}^{i} = y^{i})}{N}, \tag{5}$$

In this equation, N represents the number of samples, I is the indicator function, y_{obs}^{i} is the observed target (true intensity level), and y^{i} represents the output of the classifier (predicted intensity level).

A higher accuracy score indicates a better classification performance. An accuracy score of 1 means that all EQs were classified correctly, while a score of 0 means that none of the EQs were correctly classified.

5.2. R2 Score

The R2 score represents a statistical metric which measures the goodness-of-fit of a regression model. It estimates how much of the variance in the observed target variable can be explained by the model's independent variables.

The R2 score is calculated using the following formula:

$$R^2 = 1 - \frac{SSE}{TSS}, \tag{6}$$

where SSE (Sum of Squared Errors) represents the sum of the squared differences between the observed target values (y_{obs}^{i}) and the predicted values (y^{i}), and TSS (Total Sum of Squares) represents the sum of the squared differences between the observed target values and the mean of the observed target values ($\tilde{y}_{\text{obs}}^{i}$).

The R2 score ranges from $-\infty$ to 1. A score of 1 indicates a perfect fit, where the model explains all the variability in the observed target values. A score of 0 means that the model does not explain any of the variability, and a negative score indicates that the model performs worse than simply predicting the mean of the observed target values.

5.3. F1 Score

The F1 score is a popular metric for measuring classification model performance, especially in situations when the classes are imbalanced. It combines precision and recall to provide a balanced assessment of performance.

The F1 score is calculated using the following formula:

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}, \tag{7}$$

where precision is the ratio of true positive (TP) predictions to the total number of positive predictions, and recall is the ratio of TP predictions to the total number of actual positive instances.

The F1 score ranges from 0 to 1, with a value of 1 representing the best possible classification performance.

5.4. Kappa Score

The Kappa score is a measure of inter-rater agreement that takes into account the agreement that can occur by chance [70].

The Kappa score is calculated using the following formula:

$$\kappa = \frac{2 \times (TP \times TN - FP \times FN)}{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}, \tag{8}$$

where TN represents true negative predictions, FP represents false positive predictions, and FN represents false negative predictions.

The Kappa score ranges from -1 to 1, where a score of 1 indicates perfect agreement between the model's predictions and the true values, a score of 0 represents agreement by chance, and a negative score indicates agreement worse than chance.

5.5. Matthews Correlation Coefficient (MCC)

The MCC is a metric that quantifies the quality of binary models [71]. It takes into account true positive, true negative, FP, and FN predictions.

The MCC is calculated using the following formula:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}}, \tag{9}$$

The MCC ranges from -1 to 1, with a value of 1 indicating a perfect classifier, 0 indicating a random classifier, and -1 indicating a classifier that performs exactly opposite to the true values.

5.6. Confusion Matrix

The confusion matrix represents a table that allows for a detailed analysis of the performance of a classification model. It gives a breakdown of the predictions made by the model and the actual ground truth labels [72].

The confusion matrix can be used to calculate various performance metrics, including recall, precision, accuracy, and F1 score. Here are the definitions of true positive rate (TPR) and false positive rate (FPR):

$$\text{TPR} = \frac{TP}{TP + FN}, \tag{10}$$

$$\text{FPR} = \frac{FP}{FP + TN}. \tag{11}$$

The TPR, also known as sensitivity or recall, assesses the proportion of positive cases accurately identified by the model. The FPR calculates the proportion of negative cases mistakenly categorized as positive by the model.

The confusion matrix provides a thorough perspective of the model's performance, allowing for a more in-depth review of its strengths and weaknesses when distinguishing between classes.

6. Results and Analysis

We approached our training process with thoroughness, refraining from hastily dismissing any component that we deemed potentially ineffective. Our evaluation of intensity estimation involved assessing the suggested approach at both 3 s and 2 s following *p*-wave detection. To streamline the procedure and provide a focused demonstration of the model's efficacy, we chose to present data solely for the 2 s interval following the *p*-wave. Additionally, recognizing the variability in our continuous feature values, we implemented a normalization process. This entailed ensuring uniform weight distribution among feature samples by dividing the raw waveforms (samples) by their highest absolute value and incorporating this value into the resultant raw waveforms. Importantly, our dataset contained no missing values.

Afterward, we proceeded to partition the data frame into several split percentages to construct the training dataset, denoted as INSTANCE, and the test set. The training dataset INSTANCE encompassed 60%, 70%, or 80% of the overall feature data, while the test set comprised the remaining 40%, 30%, or 20% of the complete dataset. Subsequently, we fine-tuned the hyperparameters during the training process. To comprehensively assess the performance of our models, we employed all of the machine-learning classification

models that we had examined, calculating the error rate for each split percentage. Table 1 provides a summary of the accuracy achieved by the adopted machine-learning models, offering valuable insights into their comparative performance.

Table 1. Accuracy summary of the utilized models.

Metric	Conversion Method	XGB	DT	ET	RF	KNN
R2	[39]	0.990	0.860	0.971	0.962	0.946
	[40]	0.985	0.804	0.98	0.971	0.913
F1 score	[39]	0.973	0.824	0.929	0.910	0.892
	[40]	0.9668	0.820	0.947	0.924	0.866
MCC	[39]	0.968	0.783	0.921	0.9015	0.869
	[40]	0.9603	0.786	0.937	0.9084	0.843
Kappa	[39]	0.968	0.783	0.918	0.896	0.867
	[40]	0.960	0.786	0.937	0.9079	0.842

The adopted structure is mostly based on three main directions. The process involves selecting features, optimizing a statistical machine-learning framework, and comparing the selected machine-learning approaches with the classes that are best distinguished. The strategy aims to optimize the hyperparameters associated with specific features, like those occurring 1 s before the onset of the *p*-wave and within 2 s after. The dataset INSTANCE has been analyzed using various ML models. The classification system consists of seven intensity classifications adopted by [39] and six ones adopted by [40], denoted as $\{0, 1, \cdots, 6\}$ and $\{0, 1, \cdots, 5\}$, respectively, representing EQs from weak to extreme. It is worth mentioning that the following findings are based on the best performance obtained by an 80% to 20% split percentage.

We present the outcomes of our hyperparameter adjustment process, highlighting the pivotal discoveries that culminated in identifying the most effective model for accurately categorizing the six intensity levels. Figure 6 serves as a visual aid, showcasing the R2 score, F1 score, MCC score, and Kappa score, facilitating a straightforward comparison of the models employed. Among the models examined, the "XGB" model emerges as the frontrunner, delivering the highest level of performance. According to Figure 6a, when evaluated against the criteria set forth by [39], this model achieves impressive scores: 99.05% for R2, 97.32% for F1, and 96.83% for both Kappa and MCC. Similarly, when assessed using the metrics established by [40], as depicted in Figure 6b, the XGB model maintains its superiority with R2, F1, Kappa, and MCC scores of 98.59%, 96.7%, 96.03%, and 96.02%, respectively. These findings underscore the robustness and reliability of the XGB model in accurately categorizing EQ intensities across different evaluation criteria, thereby affirming its suitability for practical implementation in seismic hazard assessment and EQ early-warning systems.

In Figure 7, we present a comprehensive comparison of the elapsed times across all models under consideration. Notably, the "DT" model stands out with the shortest elapsed times, clocking in at 0.005 s and 0.002 s when employing methodologies outlined by [39,40], respectively. However, despite its efficiency in terms of time, the categorization performance of this model falls short of being optimal. In contrast, the XGB model, which emerges as the top performer, demonstrates a slightly longer execution time of 0.012 s according to Figure 7a, and 0.013 s as illustrated in Figure 7b, when utilizing the methodologies of [39,40], respectively. However, this marginal increase in elapsed time is justified by the significantly enhanced categorization performance it delivers. It is important to note that the other models within the accepted scheme exhibit considerably longer execution times. This trade-off between elapsed time and categorization accuracy underscores the critical need for selecting a model that strikes an appropriate balance between efficiency and performance to meet the objectives of the task at hand.

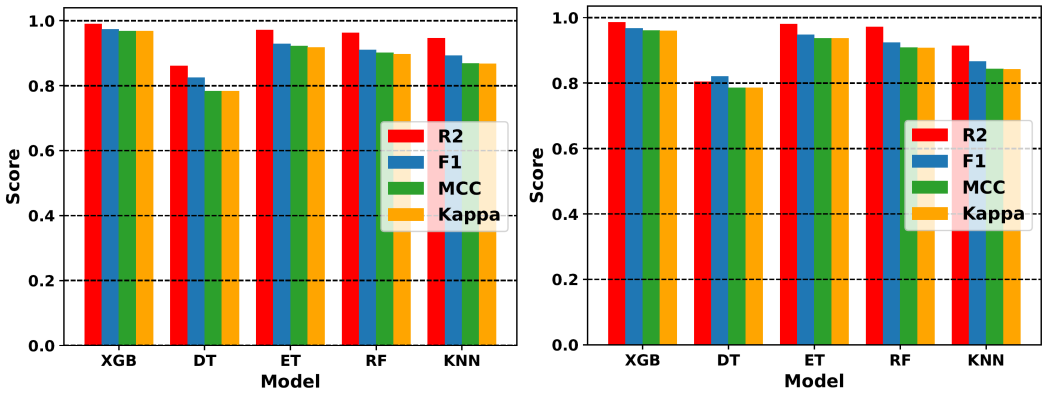


Figure 6. Accuracy of the utilized models via (**a**) equations developed by [39]; (**b**) equations developed by [40].

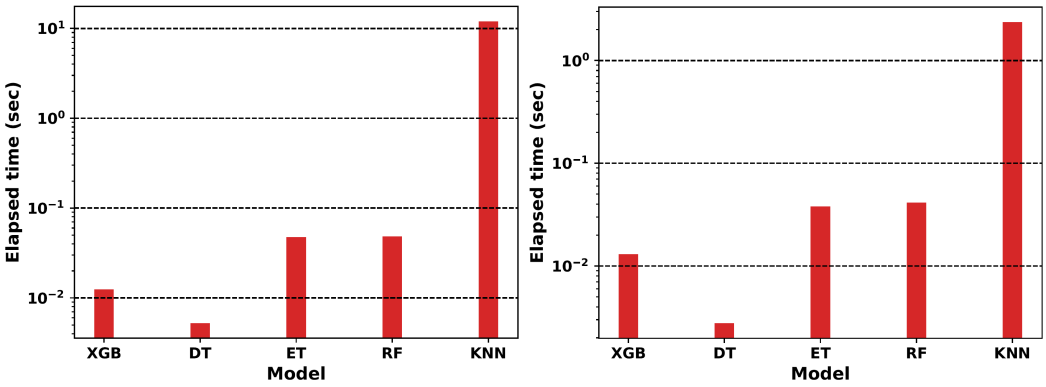
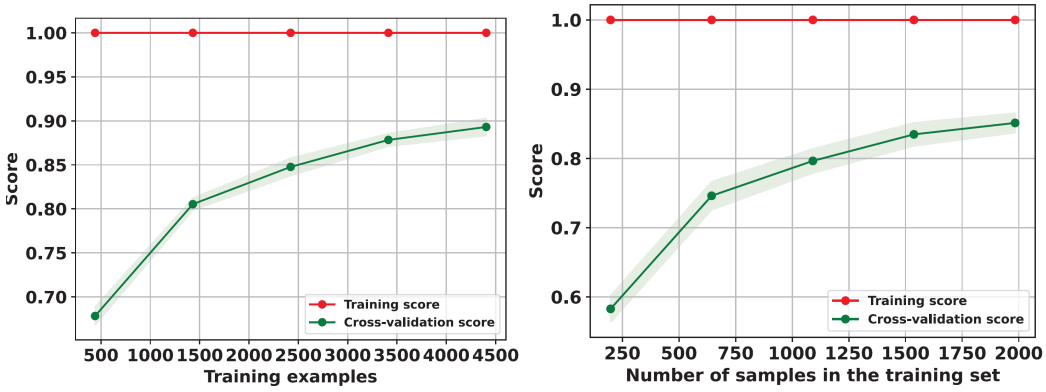


Figure 7. Elapsed time of test data via (**a**) equations developed by [39]; (**b**) equations developed by [40].

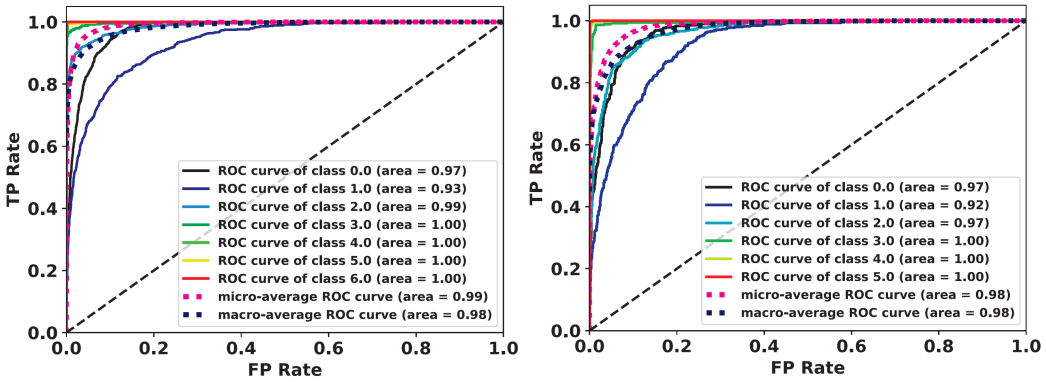
The learning curve serves as a visual representation of the effectiveness of our proposed model. In Figure 8, we observe the learning curve derived from the best-performing model, XGB. These results offer valuable insights into the model's performance in discerning EQ intensities. Upon examination, the validation curve demonstrates an approximate accuracy of 90% when utilizing the methodology outlined by [39], as depicted in Figure 8a. Similarly, employing the approach advocated by [40] yields a validation accuracy of 87%, as illustrated in Figure 8b. These findings indicate a high level of accuracy in detecting EQ intensity within the critical 2 s window following the initiation of the *p*-wave. Such accuracy is pivotal for meeting the stringent requirements of EWS, where rapid action is imperative to mitigate potential risks. Furthermore, the training and validation curves offer a compelling analogy, suggesting that the model's performance could be further refined with additional datasets (INSTANCES). This underscores the iterative nature of model development and the potential for continuous improvement to improve the reliability and accuracy of EQ intensity detection systems.

In Figure 9, we observe the ROC curve, which illustrates the relationship between the TP and FP rates, providing insights into the classification performance. Overall, our findings affirm that the XGB model excels in discrimination compared to the other benchmark models across the scenarios we investigated. In essence, the ROC curves for classifying the on-site intensity, facilitated by the superior XGB model, yield promising outcomes. These results are particularly noteworthy as they rely solely on 2 s waveform features from the initiation of the P-wave. Whether utilizing the conversion equations developed by [39] or [40], the ROC curves presented in Figure 9a and Figure 9b, respectively, underscore the efficacy of the XGB model in accurately discerning EQ intensities.



(**a**) (**b**)

Figure 8. XGB—learning curves via (**a**) equations developed by [39]; (**b**) equations developed by [40].

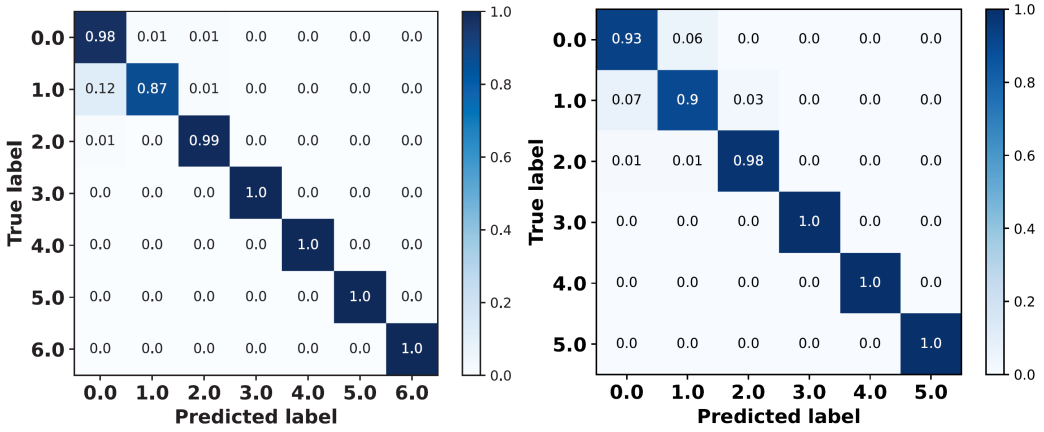


(**a**) (**b**)

Figure 9. XGB—ROC curves via (**a**) equations developed by [39]; (**b**) equations developed by [40].

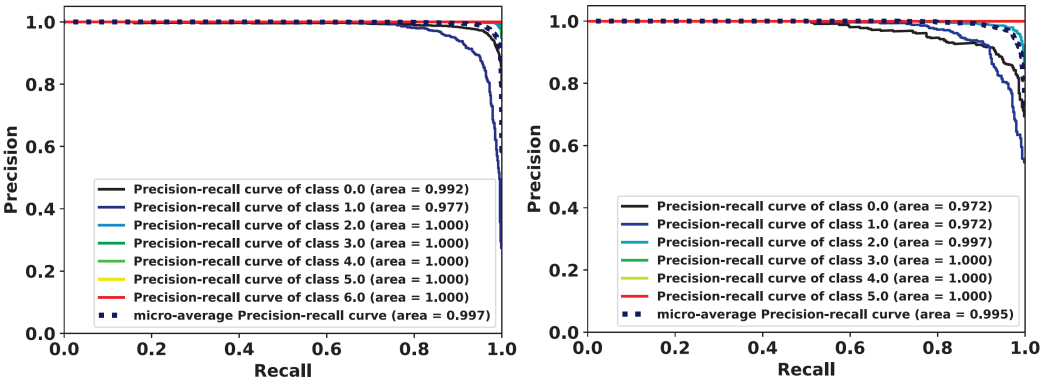
The confusion matrix serves as a pivotal tool in assessing the effectiveness of a classification task by revealing how accurately predicted classes align with actual outcomes. Examining Figure 10a,b, we observe that the "XGB" model, focusing exclusively on the 2 s waveform length from the onset of the *p*-wave, emerges as the optimal classifier for EQ intensities following our comprehensive analysis. The visual representation elucidates that, across all utilized datasets, the labels denoting very strong, severe, violent, and extreme EQs, $\{III, IV, V, VI\}$ according to [39] and $\{III, IV, V\}$ per [40], are precisely predicted with 100% accuracy. Furthermore, predictions for strong EQs exhibit a 99% accuracy rate. However, when categorizing moderate EQs, a distinction becomes more challenging. According to [39], they are classified with 87% accuracy, with a 12% misclassification rate observed between moderate and weak or light EQs (as illustrated in Figure 10a). This misclassification is expected due to the proximity in intensity among moderate, light, and weak EQs. Conversely, based on findings from [40], the XGB model achieves higher precision in categorizing moderate, weak, and light EQs, with accuracy rates of 98%, 90%, and 93%, respectively. Essentially, these results underscore the inherent complexity of delineating EQ intensities, where the distinctions among the $1, 2, \cdots, 6$ designations mirror the nuanced spectrum present in real-world seismic events.

Figure 11 depicts the precision–recall graph of our most effective machine-learning (ML) model, illustrating its classification accuracy. After extensive evaluation of various ML models, the precision–recall outcomes derived from the XGB model are portrayed, which emerge as the optimal choice utilizing 2 s waveform data from the onset of the *p*-wave. In addition, the XGB model demonstrates remarkable precision in classifying EQ intensities, ranging from 97.7% to 100%, as supported by [39] (see Figure 11a). Similarly, based on findings from [40], the XGB model achieves precision levels between 97.2% and 100%, as depicted in Figure 11b.



(**a**) (**b**)

Figure 10. XGB—confusion matrix via (**a**) equations developed by [39]; (**b**) equations developed by [40].



(**a**) (**b**)

Figure 11. XGB—precision and recall via (**a**) equations developed by [39]; (**b**) equations developed by [40].

7. Discussion

In this section, we discuss the findings of the proposed approach and show its effectiveness among the existing related ones in the literature. Indeed, PGA and intensity are among the main parameters affecting decision-making regarding disaster resilience. Several research efforts have been utilized for estimating these parameters in the literature for EWSs. In [35], the authors developed and assessed an on-site EEW system across Italy. This system uses P-wave data to measure initial peak displacement and average period (τc) in real time, predicting the intensity and alert levels. That model has been evaluated on a dataset of Italian earthquakes with a magnitude range (3.8 to 6.0); the system issued correct warnings in over 80% of cases shortly after detecting the P wave. Lead times for warnings increased with distance, offering 8–10 s at 50 km and 15–18 s at 100 km. In [57], an ANN coupled with a genetic algorithm was used to develop ground-motion models. The first model was created for various ground-motion parameters for both RotD50 and vertical components, while the second model was specifically developed for the RotD50 components of PGA, PGV, and 5% damped pseudospectral acceleration for periods between 0.01 and 3 s. These models are applicable for earthquake magnitudes ranging from 2 to 7.7 Mw and for Joyner–Boore distances from 0 to 1500 km. Another study has developed a real-time on-site estimation of response spectra based on a DL model [58]. That model was actually designed to estimate the acceleration response spectrum of expected on-site ground-motion waveforms utilizing early non-damage-causing *p*-waves and site characteristics. The model achieved 85% estimation accuracy of the ground motion. In [59], the proposed model utilized LSTM and achieved an RMSLE value of 0.804 for the maximum predicted PGA. The standard deviation error (STD) is 0.563. While the recall rate is notably high at 95.17%, the precision and F1 score are lower, at 32.4% and 48.3%, respectively. Despite this, the

false alert ratio is reasonably low at 14.7%, and the missed alert ratio is nearly zero at 0.03%. Unlike the presented efforts in the literature, our proposed model significantly outperforms previous EEW systems through several key advancements. It rapidly classifies EQ intensities within 3 s, achieving 99.05% accuracy using a dataset of 27,531 records from 386 INSN seismic stations. The model employs two distinct methodologies for converting on-site PGA to intensity, developed by the USGS and Faenza et al., ensuring comprehensive assessments. It is cost-effective, relying solely on on-site velocity waveforms, simplifying data collection, and optimizing resources while maintaining real-time accuracy. Advanced machine-learning techniques were used to identify the most effective methods for distinguishing EQ intensities, enhancing risk evaluation, and minimizing incorrect hazard assessments. Additionally, integration with IoT systems ensures reliable and swift alerts to decision-makers, crucial for effective EEWSs and evacuation plans.

8. Conclusions

Differentiating between EQ intensities is crucial for developing effective EEWSs, which are essential for saving lives, managing disasters, and assessing seismic risks. Our study introduces a novel methodology utilizing a diverse array of ML models, with the XGB model demonstrating superior performance by focusing on features observable 2 s after the initiation of the *p*-wave. Through rigorous testing, this model achieved an impressive classification accuracy of 99.05% for events with on-site intensities, as recorded by numerous seismic sensors within our research domain. This level of accuracy underscores the originality and effectiveness of our approach in accurately discerning intensity levels for EEWS implementation and real hazard assessments.

We strongly advocate for stakeholders to adopt this approach due to its demonstrated precision and reliability. Additionally, our study provides a foundation for future advancements in this field. Future work will focus on enhancing the methodology by training it with low-cost, strong-motion waveforms to improve scalability and applicability. This step is particularly important for regions with limited access to high-quality seismic sensors, as it leverages data from widely deployed sensor networks to expand coverage and accuracy.

Moreover, ensuring the practicality and effectiveness of EEWSs necessitates the real-time implementation and deployment of the developed models. Future research should aim to optimize the computational efficiency of these models, enabling them to process incoming seismic data streams in real time. Collaborations with government agencies, disaster management organizations, and seismological institutes will be crucial for the successful integration of EEWSs into existing infrastructure.

Finally, exploring time series analysis and extending the study area to include diverse geographic regions will provide a more comprehensive understanding of EQ intensities and further validate the robustness of our approach. By continually refining and expanding our methodology, we can contribute to more effective and widespread use of EEWSs, ultimately enhancing public safety and disaster preparedness globally.

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